

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : ...\\417-V-007\\Vs.pvdb

Date of Analysis : Jul 22, 2026 9:46am

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 600.000 mm ID	
Vessel Design Length, Tangent to Tangent	884.50 mm
Distance of Bottom Tangent above Grade	300.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	65 °C
Internal Design Pressure	0.050 MPa
External Design Temperature	65 °C
External Design Pressure	0.001 MPa
Maximum Allowable Working Pressure	0.082 MPa
External Max. Allowable Working Pressure	0.23 MPa
Shop Test Pressure	0.107 MPa
Field Test Pressure	0.107 MPa
Required Minimum Design Metal Temperature	11.4 °C
Warmest Computed Minimum Design Metal Temperature	3.0 °C
Wind Design Code	IS 875 (PART 3) Amd. 1&2 (2020)
Earthquake Design Code	IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Cover	SA-516 70	...	...	K02700	+IT
Flange	SA-516 70	...	...	K02700	+IT
Nozzle	SA-105	...	...	K03504	
Nozzle Flg	SA-105	...	...	K03504	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Leg Baseplate		...	...		
Leg Bolting	SA-193 B7	...	<= 64	G41400	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom flat head	0.054	0.00	0.082	3.0000	N/A	No
Shell	0.054	0.00	1.150	3.0000	N/A	No
Body flg 1	0.050	0.00	2.430	3.0000	N/A	No
Body flg 2	0.050	0.00	3.070	3.0000	N/A	No

Liquid Level: 500.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.860

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Wld Flat	12.0	12.0	12.0	12.0	10.1	...	0.85	0.85
Cylinder	712.0	700.0	6.0	6.0	4.5	3.3	0.85	0.85
Body Flg	814.5	98.0	72.0	72.0	71.6	71.6	0.85	0.85
Body Flg	884.5	70.0	70.0	70.0	33.9	31.5	0.85	0.85

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
LEGS	WRC-107/537	0.067	n/a	Passed
LEGS	WRC-107/537	0.044	n/a	Passed
N1 3"	WRC-297	3.274	Shell	Failed
N2 3"	WRC-297	3.062	Shell	Failed

Loads for Foundation/Support Design:

Total Wind Shear on top of all Legs	1004.	N
Total Earthquake Shear on top of all Legs	767.	N
Total Wind Moment at top of all Legs	530567.	N-mm
Total Earthquake Moment at top of all Legs	591667.	N-mm
Wind Load Case Test Moment on Support	338418.	N-mm
Earthquake Load Case Test Moment on Support	243778.	N-mm
Test Case Wind Shear on Support	331.	N
Max. Wind Shear on one Leg (top & bottom)	663.	N
Max. Earthq. Shear on one Leg (top & bottom)	507.	N
Max. Wind Moment at base of one Leg	530308.	N-mm
Max. Earthquake Moment at base of one Leg	405532.	N-mm
Max. Vertical Load (Wt. + Wind) on one Leg	3979.	N
Max. Vertical Load (Wt. + Eq.) on one Leg	4162.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	639.8	kgm
Shop Test - Fabricated + water (full)	865.8	kgm
Shipping - Fab. + rem. intls.+ shipping app.	639.8	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	639.8	kgm
Empty - Fab. + intls. + details + wghts.	639.8	kgm
Operating - Empty + operating liquid (no ca)	761.3	kgm
Field Test - Empty weight + test medium (full)	865.8	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N4 2"	0.082	...	...	OK	7.80	OK	No Calc[*]
N1 3"	1.15	...	...	OK	4.50	OK	No Calc[*]
N2 3"	1.15	...	...	OK	4.50	OK	No Calc[*]
N3 2"	3.07	...	...	OK	7.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 0.082 Nozzle : N4 2"
 Minimum MAWP Shells/Flanges : 0.082 Element : Bottom flat head
 Minimum MAPnc Shells/Flanges : 0.16 Element : Bottom flat head

Computed Vessel M.A.W.P. : 0.082 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
10	N4 2"	0.000	0.000	113.600
20	N1 3"	562.000	180.000	164.400
20	N2 3"	562.000	0.000	164.400
40	N3 2"	0.000	0.000	211.724

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

Checking Multiple Nozzles on Flat Head per ASME Sec. VIII Div. 1 UG-39

Comparing Nozzles on Element: Bottom flat head

Note: No Nozzle pairs found on this element.

UG-39 Nozzle Diameter and Distance to Edge Checks:

Nozzle Description	Nozzle dia. mm	Head Dia. /2 mm	Distance from Edge mm	Nozzle dia./4 mm
N4 2"	56.8	310	271.14	14.2

Comparing Nozzles on Element: Body flg 2

Note: No Nozzle pairs found on this element.



UG-39 Nozzle Diameter and Distance to Edge Checks:

Nozzle Description	Nozzle dia. mm	Head Dia. /2 mm	Distance from Edge mm	Nozzle dia./4 mm
N3 2"	56.8	319.88	339.44	14.2

No Multiple Nozzle spacing violations detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N4 2"	2.000 in	Actual	LWN	3.060	13.462	...	...	221.60	150
N3 2"	2.000 in	Actual	LWN	3.060	13.462	...	...	279.60	150
N1 3"	3.000 in	Actual	LWN	4.240	15.748	...	...	220.87	300
N2 3"	3.000 in	Actual	LWN	4.240	15.748	...	...	220.87	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N4 2"	SA-105	10.000	10.000	...	...	...
N3 2"	SA-105	70.000	10.000	...	...	...
N1 3"	SA-105	6.000	10.000	...	...	...
N2 3"	SA-105	6.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N4 2"	...	0.0	209.55	0.00	Bottom flat head
N3 2"	...	0.0	209.55	0.00	Body flg 2
N1 3"	562.000	180.0	210.00	0.00	Shell
N2 3"	562.000	0.0	210.00	0.00	Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	FLAT HEAD: 12.0mm THK X 620.0mm OD	SA-516 70
1	WELD NECK FLANGE: 72.0mm THK X 756.6mm OD	SA-516 70
1	BLIND FLANGE: 70.0mm THK X 756.6mm OD	SA-516 70
3	LEGS, SHAPE: ISA75X75X8 800.0mm	...
3	BASEPLATES: 150.0mm X 150.0mm X 14.0mm	SA-516 70
3	BASEPLATE BOLTS: 20.0mm	SA-193 B7
3	LEG RE-PADS: 150.000mm X 150.000mm X 10.000mm	SA-516 70
1	GASKET: 652.6mm OD X 626.6mm ID	...
20	BODY FLANGE BOLTS: 25.4mm DIA	SA-193 B7
40	NUTS FOR BODY FLANGE BOLTS: 25.4mm DIA	...
2	CYLINDRICAL SEGMENT 6.0mm THK X 750.0mm X 700.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 6.0mm THK X 384.9mm X 700.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Body flg 1	[11]	!	11	-104		0.027	10.000	1.00	No
Body flg 2	[11]	!	11	-104		0.027	17.500	1.00	No
Bottom flat h	[12]	!	11	3		0.850	6.000	0.85	No
Shell	[8]	!	11	-33		0.425	6.000	0.85	No
N4 2"	[1]	A	-3	-29	-29	0.724	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1 3"	[1]	!	11	-104		0.037	6.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N2 3"	[1]	!	11	-104		0.037	6.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3 2"	[1]	A	1	-104		0.001	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Bolting	[21]		-48						

Warmest MDMT: 11 3

Required Minimum Design Metal Temperature 11.4 °C

Warmest Computed Minimum Design Metal Temperature 3.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

- Impact test temps were not entered in and not considered in the analysis.
- UCS-66(i) applies to impact tested materials not by specification and
- UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
- The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Please insure the C factor entered is in accordance with Figure UG-34.

Warning: Flat head diameter not consistent with attached element.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	0.05	MPa
Design Internal Temperature	65.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	11.4	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.081
0.050	0.123
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.104
0.800	0.088
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000



Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	1.500
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.081
0.050	0.123
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117



0.670	0.104
0.800	0.088
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/4
Element From Node	10
Element To Node	20
Element Type	Flat
Description	Bottom flat head
Distance "FROM" to "TO"	12 mm
Element Outside Diameter	620 mm
Element Thickness	12 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.05 MPa
Design Temperature Internal Pressure	65 °C
Design External Pressure	0.0005 MPa
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	246 MPa
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Flat Head Attachment Factor	0.3000001



Small diameter if Non-Circular	0	mm
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Nozzle	
Detail ID	N4 2"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	76.444	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element#	2/4	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	Shell	
Distance "FROM" to "TO"	700	mm
Inside Diameter	600	mm
Element Thickness	6	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	6	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.05	MPa
Design Temperature Internal Pressure	65	°C
Design External Pressure	0.0005	MPa
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	500	mm
Liquid Density	0.0008586	kg/cm³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N1 3"	
Dist. from "FROM" Node / Offset dist	550	mm
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	156.03	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N2 3"	
Dist. from "FROM" Node / Offset dist	550	mm
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	156.03	N
Grade of Attached Flange	GR 1.1	



Nozzle Matl SA-105 +IT

Element From Node 20
 Detail Type Leg
 Detail ID LEGS
 Dist. from "FROM" Node / Offset dist 150 mm
 Diameter at Leg Centerline 669.55 mm
 Leg Orientation 3
 Number of Legs 3
 Section Identifier ISA75X75X8
 Length of Legs 800 mm

Element From Node 20
 Detail Type Weight
 Detail ID Drain pipe elbo
 Dist. from "FROM" Node / Offset dist 0 mm
 Miscellaneous Weight 98.06 N
 Offset from Element Centerline 510 mm

Element From Node 20
 Detail Type Weight
 Detail ID Dip pipe wt
 Dist. from "FROM" Node / Offset dist 550 mm
 Miscellaneous Weight 147.09 N
 Offset from Element Centerline 150 mm

Element From Node 20
 Detail Type Weight
 Detail ID Lifting lug 1
 Dist. from "FROM" Node / Offset dist 650 mm
 Miscellaneous Weight 98.06 N
 Offset from Element Centerline 300 mm

Element From Node 20
 Detail Type Weight
 Detail ID Lifting lug 2
 Dist. from "FROM" Node / Offset dist 650 mm
 Miscellaneous Weight 98.06 N
 Offset from Element Centerline 300 mm

Element# 3/4
 Element From Node 30
 Element To Node 40
 Element Type Flange
 Description Body flg 1
 Distance "FROM" to "TO" 98 mm
 Flange Inside Diameter 600 mm
 Element Thickness 72 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 72 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 0.05 MPa
 Design Temperature Internal Pressure 65 °C
 Design External Pressure 0.0005 MPa
 Design Temperature External Pressure 65 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +IT
 Perform Flange Stress Calculation (Y/N) Y
 Weld is pre-Heated No

Element# 4/4
 Element From Node 40
 Element To Node 50
 Element Type Flange
 Description Body flg 2
 Distance "FROM" to "TO" 70 mm



Flange Outside Diameter	756.6	mm
Element Thickness	70	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	70	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.05	MPa
Design Temperature Internal Pressure	65	°C
Design External Pressure	0.0005	MPa
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516	70 +IT
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N3 2"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	88.838	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Weight	
Detail ID	Pipe davit	
Dist. from "FROM" Node / Offset dist	10	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	350	mm

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Bottom flat hea		...	12	...	12
Shell		...	712	...	700
Body flg 1		...	814.5	...	98
Body flg 2		...	884.5	...	70

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

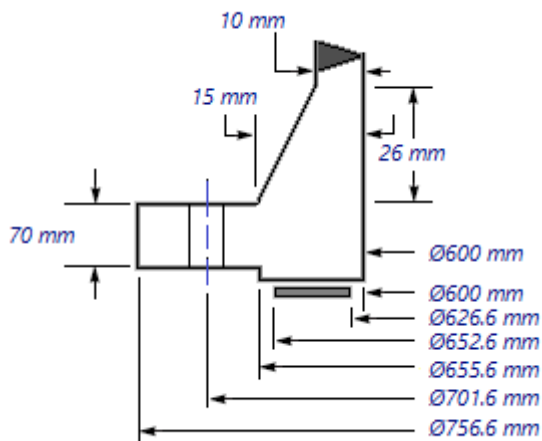
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Flange Input Data Values

Description: FLANGE-1 :

Body flg 1

Flange Type	Integral Weld Neck		
Design Pressure	P	0.05	MPa
Design Temperature		65	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	600.000	mm
Flange Outside Diameter	A	756.600	mm
Flange Thickness	t	72.0000	mm
Thickness of Hub at Small End	go	10.0000	mm
Thickness of Hub at Large End	gl	15.0000	mm
Length of Hub	h	26.0000	mm
Flange Material		SA-516 70	+IT
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	701.600	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		20	
Flange Face Outside Diameter	Fod	655.600	mm
Flange Face Inside Diameter	Fid	600.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	652.600	mm
Gasket Inside Diameter	Gi	626.600	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.05 \cdot (600/2 + 3)) / (138 \cdot 0.85 - 0.6 \cdot 0.05) + Ca$$

= 3.1292 mm

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 7) / (303 + 0.6 \cdot 7)$$

$$= 2.673 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 10) / (300 + 0.6 \cdot 10)$$

$$= 3.833 \text{ MPa}$$

The hub small end thickness is now taken as the attached shell thickness.

Corroded Flange Thickness, $t_c = T - c_i$	69.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	606.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	12.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	3.000 mm
Code R Dimension, $R = ((C - B_{cor}) / 2) - g1_{cor}$	35.800 mm
Gasket Contact Width, $N = (G_o - G_i) / 2$	13.000 mm
Basic Gasket Width, $b_o = N / 2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	639.751 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 640^2 \cdot 0.05$$

$$= 16071.068 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 640 \cdot 3 \cdot 0.05$$

$$= 3873.368 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 606^2 \cdot 0.05$$

$$= 14420.100 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 16071 - 14420$$

$$= 1650.968 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(16071 + 3873 + 0, 0)$$

$$= 19944.436 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 640 + 0 \cdot 0 \cdot 0$$

$$= 890229.062 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((19944 + 0 + 4 \cdot 0 / 640) / 172, 890229 / 172)$$

$$= 5176.284 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 69 / (3 + 0.5)$$

$$= 169.086 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 702 \cdot \sin(\pi / 20)$$

= 109.754 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max\left(\sqrt{\frac{Bs}{(2a + t)}}, 1\right)$$

$$= \max\left(\sqrt{\frac{110}{(2 \cdot 25.4 + 69)}}, 1\right)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	5176.284	7109.663	
Radial Distance between Hub and Bolts:	34.925	35.800	
Radial Distance between Bolts and Edge:	26.987	27.500	
Circ. Spacing between the Bolts:	57.150	109.754	169.086

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 7110 \cdot 172 / (68.9 \cdot \pi(653 + 627))$$

$$= 4.413 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 7110$$

$$= 1222736.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 19944 - 16071$$

$$= 3873.37 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (702 - 640) / 2$$

$$= 30.9246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((702 - 606) / 2 + 30.9)$$

$$= 39.3623 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (702 - 606 - 12) / 2$$

$$= 41.8000 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	14420.	41.8000	1.0000	603004.
Face Pressure, Mt	1651.	39.3623	1.0000	65012.
Gasket Load, Mg	3873.	30.9246	1.0000	119831.
Gasket Seating, Matm	1222736.	30.9246	1.0000	37827912.

Total Moment for Operation, Mo 787847. N-mm
 Total Moment for Gasket seating, Mg 37827912. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 42.638 mm
 Hub Ratio, h/h0 = HL / H0 0.610
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 4.000

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.772
 Factor V per 4.16.5 0.103
 Factor f per 4.16.5 4.074

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 757/606$$

$$= 1.249$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.25^2 \cdot \log_{10}(1.25))) / (1.25^2 - 1)) / (1.25 - 1)$$

$$= 8.876$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.25^2(1 + 8.55246 \cdot \log_{10}(1.25)) - 1) / ((1.0472 + 1.9448 \cdot 1.25^2) \cdot (1.25 - 1))$$

$$= 1.819$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.25^2(1 + 8.55246 \cdot \log_{10}(1.25)) - 1) / (1.36136(1.25^2 - 1) \cdot (1.25 - 1))$$

$$= 9.753$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.25^2 + 1) / (1.25^2 - 1)$$

$$= 4.579$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 9.75 \cdot 3^2 \cdot 42.6 / 0.1$$

$$= 36290.496 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.77/42.6$$

$$= 0.018 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (69 \cdot 0.018 + 1) / 1.82 + 69^3/36290$$

$$= 10.289$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (4.07 \cdot 787847/606) / (10.3 \cdot 12^2)$$

$$= 3.57 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (4.07 \cdot 37827912/606) / (10.3 \cdot 12^2)$$

$$= 171.58 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.67 \cdot 787847/606) / (10.3 \cdot 69^2)$$

$$= 0.07 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.67 \cdot 37827912/606) / (10.3 \cdot 69^2)$$

$$= 3.40 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (8.88 \cdot 787847 / (69^2 \cdot 606)) - 4.58 \cdot 0.071$$

$$= 2.10 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (8.88 \cdot 37827912 / (69^2 \cdot 606)) - 4.58 \cdot 3.4$$

= 100.78 MPa

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (3.57 + \max(0.071, 2.1)) / 2$$

$$= 2.84 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (172 + \max(3.4, 101)) / 2$$

$$= 136.18 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 19944 / 7110$$

$$= 2.81 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (890229 / 7110)$$

$$= 125.22 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	3.57	207.00	171.58	207.00
Radial Flange	0.07	138.00	3.40	138.00
Tangential Flange	2.10	138.00	100.78	138.00
Maximum Average	2.84	138.00	136.18	138.00
Bolting	2.81	172.00	125.22	172.00

Minimum Required Flange Thickness 71.628 mm

Estimated M.A.W.P. (Operating) 2.43 MPa
 Estimated Finished Weight of Flange at given Thk. 97.2 kgm
 Estimated Unfinished Weight of Forging at given Thk 126.7 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 37827912 / 1 \cdot 1 \cdot 0.1 / (10.1 \cdot 202713 \cdot 3^2 \cdot 42.6 \cdot 0.3)$$

$$= 0.860 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 787847 / 1 \cdot 1 \cdot 0.1 / (10.1 \cdot 200199 \cdot 3^2 \cdot 42.6 \cdot 0.3)$$

$$= 0.018 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 37827912 / 1 \cdot 1 \cdot 0.1 / (10.3 \cdot 202713 \cdot 3^2 \cdot 42.6 \cdot 0.3)$$

$$= 0.846 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 787847 / 1 \cdot 1 \cdot 0.1 / (10.3 \cdot 200199 \cdot 3^2 \cdot 42.6 \cdot 0.3)$$

$$= 0.017 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:



Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	2.85	207.00	133.12 207.00
Radial Flange	0.12	138.00	5.72 138.00
Tangential Flange	1.68	138.00	78.29 138.00
Maximum Average	2.27	138.00	105.70 138.00
Bolting	2.81	172.00	125.22 172.00

Minimum Req'd Thickness, ambient, no Corrosion: 62.992 mm

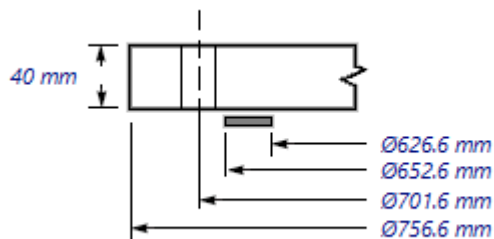
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Flange Input Data Values

Description: FLANGE-2 :

Body flg 2

Flange Type		Blind	
Design Pressure	P	0.05	MPa
Design Temperature		65	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	756.600	mm
Flange Thickness	t	70.0000	mm
Flange Material		SA-516 70 +IT	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Perimeter along the Center of the Bolts	L	2204.141	mm
Diameter of Bolt Circle	C	701.600	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		20	
Flange Face Outside Diameter	Fod	655.600	mm
Flange Face Inside Diameter	Fid	600.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	652.600	mm
Gasket Inside Diameter	Gi	626.600	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = C_b \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	639.751 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 640^2 \cdot 0.05$
 $= 16071.068 \text{ N}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 640 \cdot 3 \cdot 0.05$$

$$= 3873.368 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(16071 + 3873 + 0, 0)$$

$$= 19944.436 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 640 + 0 \cdot 0$$

$$= 890229.062 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((19944 + 0 + 4 \cdot 0/640)/172, 890229/172)$$

$$= 5176.284 \text{ mm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	5176.284	7109.663	
Radial Distance between Bolts and Edge:	26.987	27.500	
Circ. Spacing between the Bolts:	57.150	109.754	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 7110 \cdot 172 / (68.9 \cdot \pi (653 + 627))$$

$$= 4.413 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 7110$$

$$= 1222736.00 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 19944.44 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (702 - 640) / 2$$

$$= 30.9246 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 19944 \cdot 30.9 \cdot 1 / (67^2 \cdot 640) + 0.3 \cdot 1 \cdot 0.05 \cdot 640^2 / 67^2$$

$$= 1.78 \text{ MPa}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1222736 \cdot 30.9 \cdot 1 / (67^2 \cdot 640)$$

$$= 25.02 \text{ MPa}$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 19944 / 7110$$

$$= 2.81 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (890229 / 7110)$$



= 125.22 MPa

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	1.78	138.00	25.02	138.00
Bolting	2.81	172.00	125.22	172.00

Reqd. Blind Flange Thickness at Center 33.943 mm
 Reqd. Blind Flange Thickness at Gasket 33.943 mm

Estimated M.A.W.P. (Operating) 3.07 MPa
 Estimated Finished Weight of Flange at given Thk. 243.9 kgm
 Estimated Unfinished Weight of Forging at given Thk 243.9 kgm

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	1.63	138.00	22.92	138.00
Bolting	2.81	172.00	125.22	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 28.550 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Bottom flat head		0.05422	12	3	620	117.3	1
Shell		0.05422	6	3	600	117.3	2
Body flg 1		0.05	72	3	600	117.3	3
Body flg 2		0.05	70	3	756.6	117.3	4

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Bottom flat head		0.05	0.082	0.16	12	10.1359	1
Shell		0.05	1.15	2.32	6	4.5	2
Body flg 1		0.05	2.43	3.04	72	71.628	3
Body flg 2		0.05	3.07	3.07	70	33.9431	4

Minimum 0.082 0.16

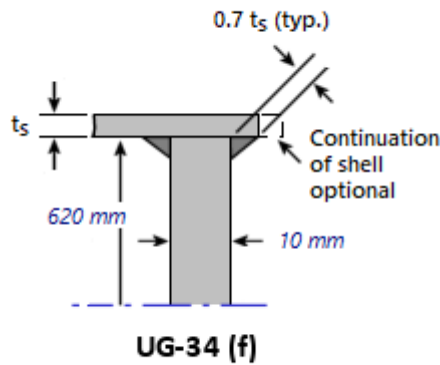
MAWP: 0.082 MPa, limited by: Bottom flat head.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom flat head

Flat Head Design Information:					
Design Pressure	P	0.054 MPa	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		65 °C
Length		12.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Circular Diameter		620.000 mm	Actual Stress		86.8 MPa
Spec. Min. Thk t _e , or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		0.0 mm ²	SG of Contents		0.000
Element Volume		0.0 ltr	Weight of Contents		0.0 N
Empty Weight		275.3 N	Operating Weight		275.3 N
			Comp. All. Ope. Stress		137.4 MPa
			Attachment Factor	C	0.30

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= d \cdot \sqrt{(Z \cdot C \cdot P / (S \cdot E))} \text{ per UG-34 (c)(3)}$$

$$= 606 \cdot \sqrt{(1 \cdot 0.3 \cdot 0.054 / (138 \cdot 0.85))}$$

$$= 7.1359 + 3.0000 = 10.1359 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.004 MPa

$$= (t/d)^2 \cdot ((S \cdot E) / (C \cdot Z)) \text{ UG-34 (c)(3)}$$

$$= (9/606)^2 \cdot ((138 \cdot 0.85) / (0.3 \cdot 1))$$

$$= 0.086 - 0.0042 = 0.082 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (t/d)^2 \cdot ((S \cdot E) / (C \cdot Z)) \text{ per UG-34 (c)(3)}$$

$$= (12/600)^2 \cdot ((138 \cdot 0.85) / (0.3 \cdot 1))$$

$$= 0.16 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (Z \cdot C \cdot P) / (((t/d)^2) \cdot E)$$



$$\begin{aligned} &= (1 \cdot 0.3 \cdot 0.054) / (((9/606)^2) \cdot 0.85) \\ &= 86.753 \text{ MPa} \end{aligned}$$

Minimum Design Metal Temperature Results:

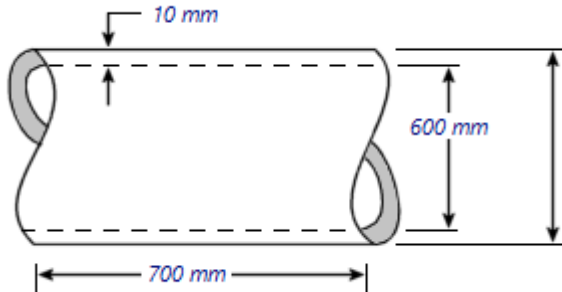
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	0.054 MPa	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		65 °C
Length		700.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	600.000 mm	Actual Stress		6.5 MPa
Spec. Min. Thk t _e , or s _a		6.0000 mm	Spec. Nominal Thk.		6.000 mm
Surface Area		1345858.2 mm ²	SG of Contents		0.860
Element Volume		198.0 ltr	Weight of Contents		1191.7 N
Empty Weight		607.7 N	Operating Weight		1799.4 N
			Comp. All. Ope. Stress		109.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.054 \cdot 303) / (138 \cdot 0.85 - 0.6 \cdot 0.054)$$

$$= 0.1401 + 3.0000 = 3.1401 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.004 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 3) / (303 + 0.6 \cdot 3)$$

$$= 1.15 - 0.0042 = 1.15 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 6) / (300 + 0.6 \cdot 6)$$

$$= 2.32 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.054 \cdot (303 + 0.6 \cdot 3)) / (0.85 \cdot 3)$$

$$= 6.480 \text{ MPa}$$



% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 0.990 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 65 °C

Elastic Modulus from Chart: CS-2 at 65 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.000	612.00	798.00	204.00	1.3039	0.0003460	34.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*34.6)/(3*204) = 0.23 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
0.260	612.00	798.00	2349.94	1.3039	0.0000088	0.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*0.88)/(3*2350) = 0.0005 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.000	612.00	16969.25	204.00	27.7275	0.0000270	2.70

$$MAEP = (4*B)/(3*(Do/t)) = (4*2.7)/(3*204) = 0.018 \text{ MPa}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.107 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.065 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.196 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.090 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.072 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.072 MPa
App 27-4	= M.A.W.P.	0.082 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 0.082 • 1
 = 0.107 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom flat head	101.4	234.0	0.434	0.11
Shell	6.7	234.0	0.029	0.11

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4 2"	Nozzle	138.00	138.00	1.000
N1 3"	Nozzle	138.00	138.00	1.000
N2 3"	Nozzle	138.00	138.00	1.000
N3 2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom flat head	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Body flg 1	138.00	138.00	1.000
Body flg 2	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N4 2"	0.28	225.00	0.001
N1 3"	0.34	225.00	0.002
N2 3"	0.34	225.00	0.002
N3 2"	0.28	225.00	0.001

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.107 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.065 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.195 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.090 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.072 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.072 MPa
App 27-4	= M.A.W.P.	0.082 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 0.082 • 1
 = 0.107 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom flat head	184.4	234.0	0.788	0.12
Shell	13.8	234.0	0.059	0.12

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4 2"	Nozzle	138.00	138.00	1.000
N1 3"	Nozzle	138.00	138.00	1.000
N2 3"	Nozzle	138.00	138.00	1.000
N3 2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom flat head	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Body flg 1	138.00	138.00	1.000
Body flg 2	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N4 2"	0.29	225.00	0.001
N1 3"	0.35	225.00	0.002
N2 3"	0.35	225.00	0.002
N3 2"	0.27	225.00	0.001

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	...	9	No Calc	No Calc
20	30	798	612	3	0.000346	34.5924
30	40	No Calc	...	69	No Calc	No Calc
40	50	No Calc	...	67	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	12	No Calc	0.0005	No Calc
20	30	6	3.26043	0.0005	0.23
30	40	72	71.628	0.0005	No Calc
40	50	70	31.5468	0.0005	No Calc

Minimum

0.23

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	798	16969.2	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	28.0792	...	21.0594	...	1.96554
20	30	61.973	197.956	31.1398	201.935	4.33811
30	40	97.9569	28.1231	93.6402	28.2709	6.85698
40	50	243.922	...	243.922	...	17.0745
Total		431	226.08	389	230.21	30

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Nozl	8.34136	...	116.775	...	N4 2"
20	Liqd	121.524	...	250	...	Liquid: 20
20	Nozl	17.0259	-405.004	550	...	N1 3"
20	Nozl	17.0259	405.004	550	...	N2 3"
20	Legs	30.5708	...	-250	...	LEGS
20	Wght	10	510	...	...	Drain pipe elbo
20	Wght	15	150	550	...	Dip pipe wt
20	Wght	10	300	650	...	Lifting lug 1
20	Wght	10	-300	650	...	Lifting lug 2
40	Nozl	9.69368	...	174.775	...	N3 2"
40	Wght	50	350	9.99997	...	Pipe davit

Total Weight of Each Detail Type:

Liquid	121.5
Nozzles	52.1
Legs	30.6
Weights	95.0
Sum of the Detail Weights	299.2 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	462.2	462.2	462.2	462.2	462.2	462.2
Nozzles	52.1	52.1	52.1	52.1	52.1	52.1
Legs	30.6	30.6	30.6	30.6	30.6	30.6
Wld Weights	95.0	95.0	95.0	95.0	95.0	95.0
Ope. Liquid	...	...	...	...	...	121.5
Test Liquid	...	225.9	...	...	...	...
Totals	639.8	865.8	639.8	639.8	639.8	761.3

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	639.8 kgm
Shop Test Wt.	- Fabricated weight + water (full)	865.8 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	639.8 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	639.8 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	639.8 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	761.3 kgm
Field Test Wt.	- Empty weight + test medium (full)	865.8 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating) 446.7 kgm
 Mass of the Upper 1/3 of the Vertical Vessel (filled w/water) 508.9 kgm
 Mass of the Upper 1/3 of the Vertical Vessel (empty) 457.1 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	...	...
20	30	1345858	1319469
30	40	389451	184726
40	50	615981	...

Total 2351290.2 1504194.6 mm²
 Total 2.4 1.5 m²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	38.3861	38.3861	38.3861	...	38.3861
20	Legs	31.1492	57.19	73.5424	...	31.1492
Legs	30	114.214	209.697	269.655	...	114.214
30	40	104.814	104.814	132.92	...	104.814
40	50	320.69	320.69	320.69	171675	320.69

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	...	...	...
20	Legs	-38.3861	-38.3861	-38.3861
Legs	30	539.718	635.201	723.265
30	40	425.504	425.504	453.61
40	50	320.69	320.69	320.69

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	...	...	...
20	Legs	...	...	...
Legs	30	171675	171675	171675
30	40	171675	171675	171675
40	50	171675	171675	171675

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N4 2"	1.87	1.96	65	150	GR 1.1	...	...	...	...
N1 3"	4.89	5.10	65	300	GR 1.1	4.323	3.009	4.887	3.009
N2 3"	4.89	5.10	65	300	GR 1.1	4.219	3.112	4.887	3.112
N3 2"	1.87	1.96	65	150	GR 1.1	...	...	...	...
Min Rating	1.868	1.960, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1 3"	111.125	0.50	1081665	3000	0.082	3.009	Pass
N2 3"	111.125	0.50	1081665	2000	0.082	3.112	Pass

Evaluating Flange on Nozzle [N1 3"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1081665) + 4 \cdot \text{abs}(3000)111.125 \leq \pi \cdot 111.125^3 \cdot [(4.887 - 0.082) + .500 \cdot 4.887]$$

$$= 0.18641\text{E}+08 \leq 0.31261\text{E}+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1 3" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.887, 0.500 \cdot 4.887 + 4.887 - 18640648.0 / (\pi \cdot 111.125^3))$$

$$= \min(4.887, 3.009)$$

$$= 3.009 \text{ MPa}$$

Evaluating Flange on Nozzle [N2 3"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1081665) + 4 \cdot \text{abs}(2000)111.125 \leq \pi \cdot 111.125^3 \cdot [(4.887 - 0.082) + .500 \cdot 4.887]$$

$$= 0.18196\text{E}+08 \leq 0.31261\text{E}+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2 3" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.887, 0.500 \cdot 4.887 + 4.887 - 18196008.0 / (\pi \cdot 111.125^3))$$

$$= \min(4.887, 3.112)$$

$$= 3.112 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 38.5614 Hz.

The Natural Frequency for the Vessel (Ope...) is 36.5692 Hz.

The Natural Frequency for the Vessel (Filled) is 34.725 Hz.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Height to Diameter ratio: 1.429

Checking the requirement for dynamic effect of wind (9.1):

Vessel Operating Natural Frequency: 36.569 Hz

Note: Vessel Natural frequency is ≥ 1 Hz.

Note: Vessel height to diameter ratio (1.429) is ≤ 5 .

Note: Dynamic wind analysis is not required.

Design Wind Speed [Vz]:

$$= \text{Basic Wind Speed} \cdot k_1 \cdot k_2 \cdot k_3 \cdot k_4 \quad (\text{for the 1st element})$$

$$= 50 \cdot 1 \cdot 1 \cdot 1 \cdot 1.15$$

$$= 57.496 \text{ meters/s}$$

Design Wind Pressure per 4.5.2 [pd]:

$$= \max(0.7 \cdot p_z, K_d \cdot K_a \cdot K_c \cdot 0.6 \cdot V_z^2)$$

$$= \max(0.7 \cdot 1984, 1 \cdot 1 \cdot 1 \cdot 0.6 \cdot 57.5^2)$$

$$= 1983.5 \text{ N/m}^2$$

Element Wind Load (operating condition):

$$= \text{Wind Area} \cdot C_f \cdot p_d$$

$$= 0.0089 \cdot 0.7 \cdot 1984$$

$$= 12.4 \text{ N}$$

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	306.00	1.0000	1.0000	1.0000	57.50	0.7000
20	662.00	1.0000	1.0000	1.0000	57.50	0.7000
30	1065.50	1.0000	1.0000	1.0000	57.50	0.7000
40	1149.50	1.0000	1.0000	1.0000	57.50	0.7000

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm



f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(104)$
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, N
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, $\leq 0.31416E-05$ (High); $0.31416E-05 < 0.39270E-05$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0/884 = 0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[2.03/2.9^2] \cdot 10^4 = 2412$ [Geometry Violation]

Compute the vibration possibility. If $V_p > 0.39270E-05$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 7023 / (884 \cdot 687^2)$$

$$= 0.16821E-04$$

Since V_p is $> 0.39270E-05$ no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Drain pipe el	0.00	312.00	1.98	0.00
Dip pipe wt	0.00	862.00	1.98	0.00
Lifting lug 1	0.00	962.00	1.98	0.00
Lifting lug 2	0.00	962.00	1.98	0.00
Pipe davit	0.00	1124.50	1.98	0.00

The Natural Frequency for the Vessel (Ope...) is 36.5692 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	306	744	8928	1.98426	13.6357
20	30	662	734.4	514080	1.98426	785.153
30	40	1065.5	720	70560	1.98426	107.766
40	50	1149.5	907.92	63554.4	1.98426	97.0664

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2015 + 2018 Amendment 2

Input Data:

Importance Factor as Entered by User I 1.5000
 Zone Number 2
 Soil Type Rocky or Hard Soil
 Response reduction factor (Force factor) R 2.0000
 Time Period of Vibration (program or user) 0.0300 Seconds
 Value of Sa/g from user Spectrum Data 0.1020
 Percent Damping 2.0 %
 Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa / g \cdot DampVal) / (R / I)$
 $= (0.1 \cdot 1.4) / 1.33$
 $= 0.107$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.081
0.050	0.123
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.104
0.800	0.088
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.03 secs
 Computed Sa/g = 0.1

Note:

It was indicated to apply the damping factor, this assumes that the Site Spectrum specified is for 5% damping. If your Sa/g values are already per the actual damping value of the vessel then do not apply the damping factor.

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 376 \cdot 0.11 \cdot 1$
 $= 40.3 \text{ N}$

Vertical load per 7.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.11 \cdot 7166 \cdot 1$
 $= 511.7 \text{ N}$

The Natural Frequency for the Vessel (Ope...) is 36.5692 Hz.

Earthquake Load Calculation:



From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	6	376.414	40.3139	40.3139	-26.8759
20	Legs	162	560.805	60.0622	32.7136	-66.9174
Legs	30	437	2056.29	220.228	119.95	444.734
30	40	765.5	1027.8	110.078	110.078	297.916
40	50	849.5	3144.68	336.796	336.796	224.53

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	156	...	...	...	...
20	Legs	75	13.6357	40.3139	81.8474	241.982
Legs	30	275	1003.62	767.478	286790	347889
30	40	603.5	204.832	446.874	18197.8	50208
40	50	687.5	97.0664	336.796	3398.71	11792.6

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	...	0.47573	0.47573	0.00089118
20	Legs	13.6357	0.47575	0.4758	0.00089213
Legs	30	1003.62	0.47625	0.4769	0.00089366
30	40	204.832	0.47702	0.47715	0.00089366
40	50	97.0664	0.47723	0.47732	0.00089366

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	488.415	3052.59
20	30	482.113	3013.21
30	40	472.66	2954.12
40	50	596.024	3725.15

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 4.422 Actual deflection : 0.477 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	23825.8	17954.2	3695665	2810954
20	30	11422.8	5739.67	1713763	869604
30	40	11422.8	5739.67	1713763	869604
40	50	11422.8	5739.67	1713763	869604

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Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	140.757	238.676	-137.356	-137.356
20	Legs	140.757	238.676	-109.567	-109.567
Legs	30	140.757	238.676	-109.567	-109.567
30	40	140.757	238.676	-109.567	-109.567
40	50	140.757	238.676	-109.567	-109.567

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	0.85944	-0.0090712	1.83289
20	30	2.515	-0.025626	5.36359
30	40	...	...	...
40	50	...	...	...

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	...	...	...	...	...
20	Legs	0.065587	0.065587	0.065587	...	...
Legs	30	-0.92216	-0.87767	-0.92216	0.19735	0.19735
30	40	-0.72702	-0.72702	-0.72702	0.19735	0.19735
40	50	-0.54793	-0.54793	-0.54793	0.19735	0.19735

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	...	...	...	...	...
20	Legs	...	0.94090E-04	0.00027818	0.31050E-04	...
Legs	30	0.19735	0.32969	0.39993	0.1088	...
30	40	0.19735	0.02092	0.057718	0.0069035	...
40	50	0.19735	0.0039071	0.013557	0.0012893	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	...
20	Legs	...	...	...	0.00027818
Legs	30	...	...	...	0.37057
30	40	...	...	...	0.057718
40	50	...	...	...	0.013557

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	-0.001497
20	Legs	...	-0.01166
Legs	30	...	0.077491
30	40	...	0.051909

40	50	...	0.039122
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Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	Legs	...	...	...	...
Legs	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	Legs	...
Legs	30	...
30	40	...
40	50	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	137.36	0.0000	-0.0000
20	0.07	140.76		109.57	0.0005	
20		140.76	-1.45	109.57		0.0132

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.00	137.36		0.0000
20	0.05	140.76		109.57	0.0004	
20		140.76	-1.41	109.57		0.0129

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	137.36	0.0000	-0.0000
20	0.07	140.76		109.57	0.0005	
20		140.76	-1.40	109.57		0.0128

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.00	137.36		0.0000
20	0.05	140.76		109.57	0.0004	
20		140.76	-1.41	109.57		0.0129



10		140.76	-0.00	137.36		0.0000
20	0.05	140.76		109.57	0.0004	
20		140.76	-1.40	109.57		0.0128

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	238.68	0.00	137.36	0.0000	-0.0000
20	0.07	238.68		109.57	0.0003	
20		238.68	-1.23	109.57		0.0112

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	238.68	0.00	137.36	0.0000	-0.0000
20	0.07	238.68		109.57	0.0003	
20		238.68	-1.12	109.57		0.0102

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.86	140.76		137.36	0.0061	
20	2.58	140.76		109.57	0.0183	
20	2.16	140.76		109.57	0.0154	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.86	140.76		137.36	0.0061	
20	2.57	140.76		109.57	0.0183	
20	2.31	140.76		109.57	0.0164	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.01	137.36		0.0001
20	0.04	140.76		109.57	0.0003	
20		140.76	-1.43	109.57		0.0131

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.01	137.36		0.0001
20	0.03	140.76		109.57	0.0002	
20		140.76	-1.42	109.57		0.0130

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.83	238.68		137.36	0.0077	
20	5.43	238.68		109.57	0.0227	
20	4.75	238.68		109.57	0.0199	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.83	238.68		137.36	0.0077	
20	5.43	238.68		109.57	0.0227	
20	4.64	238.68		109.57	0.0194	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10	0.86	140.76		137.36	0.0061	
20	2.58	140.76		109.57	0.0183	
20	1.79	140.76		109.57	0.0127	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.86	140.76		137.36	0.0061	
20	2.55	140.76		109.57	0.0181	
20	2.07	140.76		109.57	0.0147	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.86	140.76		137.36	0.0061	
20	2.58	140.76		109.57	0.0183	
20	1.83	140.76		109.57	0.0130	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.86	140.76		137.36	0.0061	
20	2.58	140.76		109.57	0.0183	
20	1.79	140.76		109.57	0.0127	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	137.36	0.0000	-0.0000
20	0.07	140.76		109.57	0.0005	
20		140.76	-1.08	109.57		0.0098

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.86	117.30		114.46	0.0073	
20	2.57	117.30		91.31	0.0219	
20	1.91	117.30		91.31	0.0163	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		117.30	-0.01	114.46		0.0001
20	0.03	117.30		91.31	0.0002	
20		117.30	-1.02	91.31		0.0112

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.86	140.76		137.36	0.0061	
20	2.58	140.76		109.57	0.0183	
20	1.83	140.76		109.57	0.0130	

Absolute Maximum of the all of the Stress Ratio's 0.0227

Governing Element: Shell

Governing Load Case 11 : HP+HW+HI

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
---------------------	----------------------	----------------------------	-------------------------	---------------------------	-----------------	------------



Bottom flat head	1.8	...	...	...	1.8	+ Tens
Bottom flat head	1.8	...	...	...	...	- Comp
Shell	5.4	0.1	...	...	5.4	+ Tens
Shell	5.4	0.1	...	...	...	- Comp
Shell	5.4	-0.9	0.2	0.1	4.7	+ Tens
Shell	5.4	-0.9	-0.2	-0.1	...	- Comp



Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	262.000 mm
Center of Gravity of the Nozzles	570.219 mm
Center of Gravity of the Legs	-238.000 mm
Center of Gravity of the Added Weights (Operating)	663.316 mm
Center of Gravity of the Added Weights (Empty)	663.316 mm
Center of Gravity of Bare Shell New and Cold	705.669 mm
Center of Gravity of Bare Shell Corroded	744.795 mm
Vessel CG in the Operating Condition	597.426 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	643.265 mm
Vessel CG in the Test Condition	582.962 mm

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RESULTS FOR LEGS : Operating Case Description: LEGS

Legs attached to: Shell

Section Properties : Single Angle ISA75X75X8

India ISI Structural Steel Data

Overall Leg Length	800.000	mm
Effective Leg Length	Leglen 500.000	mm
Distance Leg Up Side of Vessel	150.000	mm
Number of Legs	Nleg 3	
Cross Sectional Area for ISA75X75X8	Aleg 1137.998	mm ²
Section Inertia (strong axis)	589997.000	mm ⁴
Section Inertia (weak axis)	589997.000	mm ⁴
Section Modulus (strong axis)	11000.045	mm ³
Section Modulus (weak axis)	11000.045	mm ³
Radius of Gyration (strong axis)	22.800	mm
Radius of Gyration (weak axis)	22.800	mm

Leg Orientation - Diagonal

Overturning Moment at top of Legs	591666.9	N-mm
Total Weight Load at top of Legs	W 7185.6	N
Total Shear force at top of Legs	1003.6	N
Additional force in Leg due to Bracing	Fadd 0.0	N
Occasional Load Factor	Occfac 1.333	
Effective Leg End Condition Factor	k 1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P 150.000	mm
Pad Length along Vessel Axis	C22P 150.000	mm
Pad Thickness	Tpad 10.000	mm

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (1004) (957450 / 1450184)$
 $= 662.62 \text{ N}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W / N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (7186 / 3 + (591431 / (1 \cdot 335))) / 1138$
 $= 3.66 \text{ MPa}$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (7186 / 3) / 1138$
 $= 2.10 \text{ MPa}$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	75.000	75.000	8.0000
Distance to geometric centroid:	21.400	21.400	
Arm about YY:	17.400	20.100	
Arm about ZZ:	16.100	17.400	
Leg areas:	600.00	536.00	
Geometric inertia components YY:	0.18485E+06	0.41705E+06	
Geometric inertia components ZZ:	0.43677E+06	0.16514E+06	
Geometric inertias Iy & Iz:	0.60191E+06	0.60191E+06	
Product of inertia:	0.35554E+06		
Mohrs Radius:	0.35554E+06		
Average Inertia:	0.60191E+06		

QFACT = 1.0000 FBZ = 0.16383
 Principal Axis Inertias (Z&W) = 0.24637E+06 0.95745E+06
 Angle to Principal Axis = 45.000



Distances to extreme fibers CW & CZ = 53.033 22.769
 FOB from Eq 5-5 = 3.1165
 Bending allowables Fby & Fbz = 0.16383 0.16383

Shear Center Coordinates Wo & Zo: 23.978 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.58326E-01 2.5308
 AREA, LENGTH, Kw, Kz: 1.7639 19.685 1.0000 1.0000
 H, Few, Fez, Fej: 0.64788 979.84 252.13 148.24
 Fe computed from C4-1: 140.02

Initial (Kl/r)max, & (Kl/r)equiv = 33.982 45.599
 Final (Kl/r)max, & Cc = 45.599 127.18
 Fa based on Eq 4-1 = 0.12937

Leg Bending and Axial Stresses vs. Allowables: (MPa)

	Actual	Allowable
Weak Axis Bending :	21.65	218.38
Strong Axis Bending :	12.98	218.38
Axial Compression :	3.66	172.45

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.180

AISC Unity Check : 0.180, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Ope Condition

Rectangular Attachment Parameter	C11	106.066 mm
Rectangular Attachment Parameter	C22	130.950 mm

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	3.0000 mm
Vessel Diameter	Dv	600.000 mm
Vessel Thickness	Tv	6.000 mm
Design Temperature	T1	65.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Rectangular
Parameter C11	C11	106.07 mm
Parameter C22	C22	130.95 mm
Thickness of Reinforcing Pad	Tpad	10.000 mm
Pad Parameter C11P	C11p	150.000 mm
Pad Parameter C22P	C22p	150.000 mm
Design Internal Pressure	Dp	0.050 MPa
Include Pressure Thrust		No
Vessel Centerline Direction Cosine	Vx	0.000
Vessel Centerline Direction Cosine	Vy	1.000
Vessel Centerline Direction Cosine	Vz	0.000
Nozzle Centerline Direction Cosine	Nx	1.000
Nozzle Centerline Direction Cosine	Ny	0.000
Nozzle Centerline Direction Cosine	Nz	0.000

Global Force (SUS)	Fx	0.0	N
Global Force (SUS)	Fy	2395.2	N
Global Force (SUS)	Fz	0.0	N
Global Moment (SUS)	Mx	0.0	N-mm
Global Moment (SUS)	My	0.0	N-mm
Global Moment (SUS)	Mz	68949.9	N-mm

Internal Pressure (SUS)	P	0.05	MPa
Include Pressure Thrust		No	

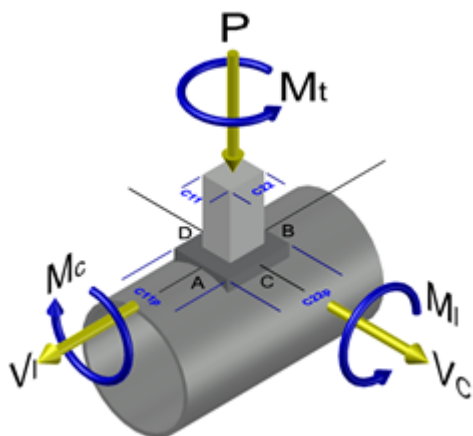
Global Force (OCC)	Fx	662.6	N
Global Force (OCC)	Fy	1766.6	N
Global Force (OCC)	Fz	0.0	N
Global Moment (OCC)	Mx	0.0	N-mm
Global Moment (OCC)	My	0.0	N-mm
Global Moment (OCC)	Mz	50855.7	N-mm

Occasional Internal Pressure (OCC)	Pvar	0.00	MPa
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Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	0.0	N
Circumferential Shear	VC	-0.0	N
Longitudinal Shear	VL	2395.2	N
Circumferential Moment	MC	-0.0	N-mm
Longitudinal Moment	ML	-68949.9	N-mm
Torsional Moment	MT	-0.0	N-mm

Dimensionless Parameters used : Gamma = 23.81

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.214	4C	3.529	(A,B)
N(PHI) / (P/Rm)	0.214	3C	2.464	(C,D)



M(PHI) / (P)	0.185	2C1	0.056	(A,B)
M(PHI) / (P)	0.185	1C	0.087	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.184	3A	0.907	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.204	1A	0.086	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.197	3B	2.668	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.199	1B	0.033	(A,B,C,D)
N(x) / (P/Rm)	0.200	3C	2.605	(A,B)
N(x) / (P/Rm)	0.200	4C	3.611	(C,D)
M(x) / (P)	0.202	1C1	0.083	(A,B)
M(x) / (P)	0.202	2C	0.049	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.184	4A	1.433	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.233	2A	0.040	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.197	4B	0.885	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.212	2B	0.051	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.7	0.7	-0.7	-0.7	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.3	-1.3	-1.3	1.3	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.0	-0.7	-2.0	0.7	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		1.9	-1.9	-1.9	1.9	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		2.2	-1.7	-2.2	1.7	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Str. Int.		2.2	1.7	2.2	1.7	1.4	1.4	1.4	1.4

Dimensionless Parameters used : Gamma = 101.50

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.246	4C	9.518	(A,B)
N(PHI) / (P/Rm)	0.246	3C	3.723	(C,D)
M(PHI) / (P)	0.246	2C1	0.011	(A,B)
M(PHI) / (P)	0.246	1C !	0.067	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.246	3A	2.778	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.246	1A	0.054	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.246	3B	5.971	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.246	1B	0.008	(A,B,C,D)
N(x) / (P/Rm)	0.246	3C	3.723	(A,B)
N(x) / (P/Rm)	0.246	4C	9.518	(C,D)
M(x) / (P)	0.246	1C1	0.030	(A,B)
M(x) / (P)	0.246	2C !	0.036	(C,D)



N(x) / (MC/(Rm**2 * Beta)) 0.246 4A 9.123 (A,B,C,D)
 M(x) / (MC/(Rm * Beta)) 0.246 2A 0.022 (A,B,C,D)
 N(x) / (ML/(Rm**2 * Beta)) 0.246 4B 3.030 (A,B,C,D)
 M(x) / (ML/(Rm * Beta)) 0.246 2B 0.011 (A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		6.0	6.0	-6.0	-6.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		4.6	-4.6	-4.6	4.6	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		10.6	1.4	-10.6	-1.4	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		3.0	3.0	-3.0	-3.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		6.5	-6.5	-6.5	6.5	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		9.5	-3.4	-9.5	3.4	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-2.7	-2.7	2.7	2.7
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-2.7	-2.7	2.7	2.7
Str. Int.		10.6	4.8	10.6	4.8	5.3	5.3	5.3	5.3

WRC 107 Stress Calculation for OCCasional loads:

Radial Load P 662.6 N
 Circumferential Shear VC -0.0 N
 Longitudinal Shear VL 1766.6 N
 Circumferential Moment MC -0.0 N-mm
 Longitudinal Moment ML -50855.7 N-mm
 Torsional Moment MT -0.0 N-mm

Dimensionless Parameters used : Gamma = 23.81
 Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.6	-0.6	-0.6	-0.6	-0.4	-0.4	-0.4	-0.4
Circ. Bend. P		-1.3	1.3	-1.3	1.3	-2.1	2.1	-2.1	2.1
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.5	0.5	-0.5	-0.5	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.0	-1.0	-1.0	1.0	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		-0.4	0.3	-3.4	1.2	-2.5	1.6	-2.5	1.6
Long. Memb. P		-0.4	-0.4	-0.4	-0.4	-0.6	-0.6	-0.6	-0.6



Long. Bend. P	-2.0	2.0	-2.0	2.0	-1.2	1.2	-1.2	1.2
Long. Memb. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC	-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML	0.2	0.2	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	1.4	-1.4	-1.4	1.4	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	-0.8	0.3	-4.0	2.7	-1.8	0.6	-1.8	0.6
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Str. Int.	0.8	0.3	4.0	2.7	2.7	1.9	2.7	1.9

Dimensionless Parameters used : Gamma = 101.50
 Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-6.9	-6.9	-6.9	-6.9	-2.7	-2.7	-2.7	-2.7
Circ. Bend. P		-4.8	4.8	-4.8	4.8	-29.4	29.4	-29.4	29.4
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		4.4	4.4	-4.4	-4.4	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		3.4	-3.4	-3.4	3.4	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		-3.9	-1.1	-19.6	-3.1	-32.1	26.7	-32.1	26.7
Long. Memb. P		-2.7	-2.7	-2.7	-2.7	-6.9	-6.9	-6.9	-6.9
Long. Bend. P		-13.2	13.2	-13.2	13.2	-16.1	16.1	-16.1	16.1
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		2.2	2.2	-2.2	-2.2	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		4.8	-4.8	-4.8	4.8	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		-8.9	8.0	-22.9	13.0	-23.0	9.1	-23.0	9.1
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-2.0	-2.0	2.0	2.0
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-2.0	-2.0	2.0	2.0
Str. Int.		8.9	9.0	22.9	16.1	32.5	26.9	32.5	26.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		1.1	1.2	1.1	1.2	1.1	1.2	1.1	1.2
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		1.1	1.2	1.1	1.2	1.1	1.2	1.1	1.2
Circ. Pl (SUS)		0.7	0.7	-0.7	-0.7	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		-0.1	-0.1	-1.1	-1.1	-0.4	-0.4	-0.4	-0.4
Circ. Pl(TOTAL)		0.6	0.6	-1.7	-1.7	-0.4	-0.4	-0.4	-0.4
Circ. Q (SUS)		1.3	-1.3	-1.3	1.3	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		-0.4	0.4	-2.3	2.3	-2.1	2.1	-2.1	2.1
Circ. Q (TOTAL)		1.0	-1.0	-3.6	3.6	-2.1	2.1	-2.1	2.1

Long. Pm (SUS)	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Long. Pl (SUS)	0.3	0.3	-0.3	-0.3	0.0	0.0	-0.0	-0.0
Long. Pl (OCC)	-0.2	-0.2	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6
Long. Pl(TOTAL)	0.0	0.0	-0.9	-0.9	-0.6	-0.6	-0.6	-0.6
Long. Q (SUS)	1.9	-1.9	-1.9	1.9	0.0	0.0	-0.0	0.0
Long. Q (OCC)	-0.6	0.6	-3.4	3.4	-1.2	1.2	-1.2	1.2
Long. Q (TOTAL)	1.3	-1.3	-5.3	5.3	-1.2	1.2	-1.2	1.2
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear Pl (OCC)	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear Pl(TOTAL)	0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	1.1	1.2	1.1	1.2	1.1	1.2	1.1	1.2
Pm (SUS+OCC)	1.1	1.2	1.1	1.2	1.1	1.2	1.1	1.2
Pm+Pl (SUS)	1.8	1.9	0.5	0.5	1.6	1.6	1.6	1.6
Pm+Pl (SUS+OCC)	1.7	1.8	0.6	0.5	2.6	2.6	2.6	2.6
Pm+Pl+Q (Total)	2.7	1.6	5.6	5.0	2.5	3.5	2.5	3.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1.19	138.00	Passed
Pm (SUS+OCC)	1.19	165.60	Passed
Pm+P1 (SUS)	1.86	207.00	Passed
Pm+P1 (SUS+OCC)	2.58	248.39	Passed
Pm+P1+0 (TOTAL)	5.56	492.02	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The P_m+P_l+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		5.0	5.1	5.0	5.1	5.0	5.1	5.0	5.1
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		5.0	5.1	5.0	5.1	5.0	5.1	5.0	5.1
Circ. Pl (SUS)		6.0	6.0	-6.0	-6.0	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		-2.5	-2.5	-11.3	-11.3	-2.7	-2.7	-2.7	-2.7
Circ. Pl(TOTAL)		3.5	3.5	-17.3	-17.3	-2.7	-2.7	-2.7	-2.7
Circ. Q (SUS)		4.6	-4.6	-4.6	4.6	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		-1.4	1.4	-8.2	8.2	-29.4	29.4	-29.4	29.4
Circ. Q (TOTAL)		3.2	-3.2	-12.9	12.9	-29.4	29.4	-29.4	29.4
Long. Pm (SUS)		2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Long. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)		2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5

Long. P1 (SUS)	3.0	3.0	-3.0	-3.0	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	-0.5	-0.5	-4.9	-4.9	-6.9	-6.9	-6.9	-6.9
Long. P1(TOTAL)	2.6	2.6	-8.0	-8.0	-6.9	-6.9	-6.9	-6.9
Long. Q (SUS)	6.5	-6.5	-6.5	6.5	0.0	0.0	-0.0	0.0
Long. Q (OCC)	-8.4	8.4	-17.9	17.9	-16.1	16.1	-16.1	16.1
Long. Q (TOTAL)	-1.9	1.9	-24.4	24.4	-16.1	16.1	-16.1	16.1
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-2.7	-2.7	2.7	2.7
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-2.0	-2.0	2.0	2.0
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-4.6	-4.6	4.6	4.6
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	5.0	5.1	5.0	5.1	5.0	5.1	5.0	5.1
Pm (SUS+OCC)	5.0	5.1	5.0	5.1	5.0	5.1	5.0	5.1
Pm+P1 (SUS)	11.0	11.1	1.0	0.9	6.7	6.7	6.7	6.7
Pm+P1 (SUS+OCC)	8.6	8.6	12.3	12.3	11.4	11.5	11.4	11.5
Pm+P1+Q (Total)	11.8	7.1	29.9	18.9	29.4	32.8	29.4	32.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.08	138.00	Passed
Pm (SUS+OCC)	5.08	165.60	Passed
Pm+P1 (SUS)	11.08	207.00	Passed
Pm+P1 (SUS+OCC)	12.32	248.39	Passed
Pm+P1+Q (TOTAL)	32.77	492.02	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-516 70
Baseplate Allowable Stress	SBA 137.90 MPa
Baseplate Length	B 150.0000 mm
Baseplate Width	D 150.0000 mm
Baseplate Thickness	BTHK 14.0000 mm
Leg Dimension Along Baseplate Length	d 74.9998 mm
Leg Dimension Along Baseplate Width	b 74.9998 mm
Bolt Circle Diameter	BCD 440.0000 mm
Bolt Material	SA-193 B7
Bolt Allowable Tensile Stress	STBA 137.30 MPa
Bolt Allowable Shear Stress	SBShear 107.87 MPa
Anchor Bolt Nominal Diameter	BOD 20.0000 mm
Number of Anchor Bolts in Tension per Leg	NB 0
Total Number of Anchors Bolt per Leg	NBT 1
Bolt Corrosion Allowance	bca 0.0000 mm
Ultimate 28-day Concrete Strength	FCPRIME 8.750 MPa

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [rb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 1004/(217 • 3)
 = 1.5 MPa. Must be less than 107.9 MPa.

Plate thickness required (1/2"; 13mm) per D. Moss



Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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RESULTS FOR LEGS : HydroTest Case Description: LEGS

Legs attached to: Shell

Section Properties : Single Angle ISA75X75X8

India ISI Structural Steel Data

Overall Leg Length	800.000	mm
Effective Leg Length	Leglen 500.000	mm
Distance Leg Up Side of Vessel	150.000	mm
Number of Legs	Nleg 3	
Cross Sectional Area for ISA75X75X8	Aleg 1137.998	mm ²
Section Inertia (strong axis)	589997.000	mm ⁴
Section Inertia (weak axis)	589997.000	mm ⁴
Section Modulus (strong axis)	11000.045	mm ³
Section Modulus (weak axis)	11000.045	mm ³
Radius of Gyration (strong axis)	22.800	mm
Radius of Gyration (weak axis)	22.800	mm

Leg Orientation - Diagonal

Overturning Moment at top of Legs	338418.4	N-mm
Total Weight Load at top of Legs	W 8209.5	N
Total Shear force at top of Legs	331.2	N
Additional force in Leg due to Bracing	Fadd 0.0	N
Occasional Load Factor	Occfac 1.333	
Effective Leg End Condition Factor	k 1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P 150.000	mm
Pad Length along Vessel Axis	C22P 150.000	mm
Pad Thickness	Tpad 10.000	mm

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (331) (957450 / 1450184)$
 $= 218.66 \text{ N}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W / N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (8210 / 3 + (338284 / (1 \cdot 335))) / 1138$
 $= 3.29 \text{ MPa}$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (8210 / 3) / 1138$
 $= 2.40 \text{ MPa}$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	75.000	75.000	8.0000
Distance to geometric centroid:	21.400	21.400	
Arm about YY:	17.400	20.100	
Arm about ZZ:	16.100	17.400	
Leg areas:	600.00	536.00	
Geometric inertia components YY:	0.18485E+06	0.41705E+06	
Geometric inertia components ZZ:	0.43677E+06	0.16514E+06	
Geometric inertias Iy & Iz:	0.60191E+06	0.60191E+06	
Product of inertia:	0.35554E+06		
Mohrs Radius:	0.35554E+06		
Average Inertia:	0.60191E+06		

QFACT = 1.0000 FBZ = 0.16383
 Principal Axis Inertias (Z&W) = 0.24637E+06 0.95745E+06
 Angle to Principal Axis = 45.000



Distances to extreme fibers CW & CZ = 53.033 22.769
 FOB from Eq 5-5 = 3.1165
 Bending allowables Fby & Fbz = 0.16383 0.16383

Shear Center Coordinates Wo & Zo: 23.978 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.58326E-01 2.5308
 AREA, LENGTH, Kw, Kz: 1.7639 19.685 1.0000 1.0000
 H, Few, Fez, Fej: 0.64788 979.84 252.13 148.24
 Fe computed from C4-1: 140.02

Initial (Kl/r)_{max}, & (Kl/r)_{equiv} = 33.982 45.599
 Final (Kl/r)_{max}, & Cc = 45.599 127.18
 Fa based on Eq 4-1 = 0.12937

Leg Bending and Axial Stresses vs. Allowables: (MPa)

	Actual	Allowable
Weak Axis Bending :	7.15	218.38
Strong Axis Bending :	4.28	218.38
Axial Compression :	3.29	172.45

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.071

AISC Unity Check : 0.071, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Test Condition

Rectangular Attachment Parameter	C11	106.066 mm
Rectangular Attachment Parameter	C22	130.950 mm

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	3.0000 mm
Vessel Diameter	Dv	600.000 mm
Vessel Thickness	Tv	6.000 mm
Design Temperature	T1	65.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	234.00 MPa
Vessel Hot S.I. Allowable	Smh	234.00 MPa
Attachment Type	Type	Rectangular
Parameter C11	C11	106.07 mm
Parameter C22	C22	130.95 mm
Thickness of Reinforcing Pad	Tpad	10.000 mm
Pad Parameter C11P	C11p	150.000 mm
Pad Parameter C22P	C22p	150.000 mm
Design Internal Pressure	Dp	0.050 MPa
Include Pressure Thrust		No
Vessel Centerline Direction Cosine	Vx	0.000
Vessel Centerline Direction Cosine	Vy	1.000
Vessel Centerline Direction Cosine	Vz	0.000
Nozzle Centerline Direction Cosine	Nx	1.000
Nozzle Centerline Direction Cosine	Ny	0.000
Nozzle Centerline Direction Cosine	Nz	0.000
Global Force (SUS)	Fx	0.0 N
Global Force (SUS)	Fy	2736.5 N

Global Force (SUS)	Fz	0.0	N
Global Moment (SUS)	Mx	0.0	N-mm
Global Moment (SUS)	My	0.0	N-mm
Global Moment (SUS)	Mz	78774.9	N-mm

Internal Pressure (SUS)	P	0.05	MPa
Include Pressure Thrust		No	

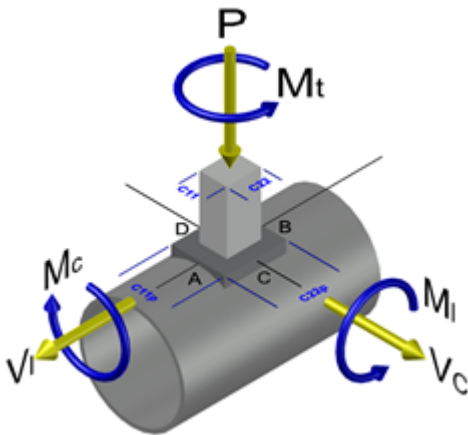
Global Force (OCC)	Fx	218.7	N
Global Force (OCC)	Fy	1010.5	N
Global Force (OCC)	Fz	0.0	N
Global Moment (OCC)	Mx	0.0	N-mm
Global Moment (OCC)	My	0.0	N-mm
Global Moment (OCC)	Mz	29088.1	N-mm

Occasional Internal Pressure (OCC)	Pvar	0.06	MPa
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Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	N
Circumferential Shear	VC	-0.0	N
Longitudinal Shear	VL	2736.5	N
Circumferential Moment	MC	-0.0	N-mm
Longitudinal Moment	ML	-78774.9	N-mm
Torsional Moment	MT	-0.0	N-mm

Dimensionless Parameters used : Gamma = 23.81

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.214	4C	3.529	(A,B)
N(PHI) / (P/Rm)	0.214	3C	2.464	(C,D)
M(PHI) / (P)	0.185	2C1	0.056	(A,B)
M(PHI) / (P)	0.185	1C	0.087	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.184	3A	0.907	(A,B,C,D)



M(PHI) / (MC/(Rm * Beta)) 0.204 1A 0.086 (A,B,C,D)
 N(PHI) / (ML/(Rm**2 * Beta)) 0.197 3B 2.668 (A,B,C,D)
 M(PHI) / (ML/(Rm * Beta)) 0.199 1B 0.033 (A,B,C,D)

 N(x) / (P/Rm) 0.200 3C 2.605 (A,B)
 N(x) / (P/Rm) 0.200 4C 3.611 (C,D)
 M(x) / (P) 0.202 1C1 0.083 (A,B)
 M(x) / (P) 0.202 2C 0.049 (C,D)
 N(x) / (MC/(Rm**2 * Beta)) 0.184 4A 1.433 (A,B,C,D)
 M(x) / (MC/(Rm * Beta)) 0.233 2A 0.040 (A,B,C,D)
 N(x) / (ML/(Rm**2 * Beta)) 0.197 4B 0.885 (A,B,C,D)
 M(x) / (ML/(Rm * Beta)) 0.212 2B 0.051 (A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.8	0.8	-0.8	-0.8	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.5	-1.5	-1.5	1.5	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.3	-0.7	-2.3	0.7	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		2.2	-2.2	-2.2	2.2	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		2.5	-1.9	-2.5	1.9	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Str. Int.		2.5	1.9	2.5	1.9	1.6	1.6	1.6	1.6

Dimensionless Parameters used : Gamma = 101.50

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.246	4C	9.518	(A,B)
N(PHI) / (P/Rm)	0.246	3C	3.723	(C,D)
M(PHI) / (P)	0.246	2C1	0.011	(A,B)
M(PHI) / (P)	0.246	1C !	0.067	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.246	3A	2.778	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.246	1A	0.054	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.246	3B	5.971	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.246	1B	0.008	(A,B,C,D)
N(x) / (P/Rm)	0.246	3C	3.723	(A,B)
N(x) / (P/Rm)	0.246	4C	9.518	(C,D)
M(x) / (P)	0.246	1C1	0.030	(A,B)
M(x) / (P)	0.246	2C !	0.036	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.246	4A	9.123	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.246	2A	0.022	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.246	4B	3.030	(A,B,C,D)



M(x) / (ML/(Rm * Beta)) 0.246 2B 0.011 (A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		6.9	6.9	-6.9	-6.9	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		5.3	-5.3	-5.3	5.3	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		12.2	1.6	-12.2	-1.6	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		3.5	3.5	-3.5	-3.5	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		7.4	-7.4	-7.4	7.4	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		10.9	-3.9	-10.9	3.9	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-3.0	-3.0	3.0	3.0
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-3.0	-3.0	3.0	3.0
Str. Int.		12.2	5.5	12.2	5.5	6.1	6.1	6.1	6.1

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	218.7	N
Circumferential Shear	VC	-0.0	N
Longitudinal Shear	VL	1010.5	N
Circumferential Moment	MC	-0.0	N-mm
Longitudinal Moment	ML	-29088.1	N-mm
Torsional Moment	MT	-0.0	N-mm

Dimensionless Parameters used : Gamma = 23.81

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.2	-0.2	-0.2	-0.2	-0.1	-0.1	-0.1	-0.1
Circ. Bend. P		-0.4	0.4	-0.4	0.4	-0.7	0.7	-0.7	0.7
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		0.6	-0.6	-0.6	0.6	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		0.2	-0.0	-1.5	0.5	-0.8	0.5	-0.8	0.5
Long. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.2	-0.2	-0.2	-0.2
Long. Bend. P		-0.6	0.6	-0.6	0.6	-0.4	0.4	-0.4	0.4
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0

Long. Memb. ML	0.1	0.1	-0.1	-0.1	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.8	-0.8	-0.8	0.8	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	0.1	-0.2	-1.7	1.2	-0.6	0.2	-0.6	0.2
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Str. Int.	0.2	0.2	1.7	1.2	1.0	0.7	1.0	0.7

Dimensionless Parameters used : $\Gamma = 101.50$
Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-2.3	-2.3	-2.3	-2.3	-0.9	-0.9	-0.9	-0.9
Circ. Bend. P		-1.6	1.6	-1.6	1.6	-9.7	9.7	-9.7	9.7
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		2.5	2.5	-2.5	-2.5	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		2.0	-2.0	-2.0	2.0	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		0.6	-0.1	-8.4	-1.3	-10.6	8.8	-10.6	8.8
Long. Memb. P		-0.9	-0.9	-0.9	-0.9	-2.3	-2.3	-2.3	-2.3
Long. Bend. P		-4.3	4.3	-4.3	4.3	-5.3	5.3	-5.3	5.3
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		1.3	1.3	-1.3	-1.3	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		2.7	-2.7	-2.7	2.7	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		-1.2	2.0	-9.2	4.9	-7.6	3.0	-7.6	3.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Str. Int.		1.9	2.1	9.2	6.2	11.0	9.0	11.0	9.0

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		1.1	1.2	1.1	1.2	1.1	1.2	1.1	1.2
Circ. Pm (OCC)		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Circ. Pm(TOTAL)		2.6	2.7	2.6	2.7	2.6	2.7	2.6	2.7
Circ. Pl (SUS)		0.8	0.8	-0.8	-0.8	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		0.1	0.1	-0.5	-0.5	-0.1	-0.1	-0.1	-0.1
Circ. Pl(TOTAL)		0.8	0.8	-1.2	-1.2	-0.1	-0.1	-0.1	-0.1
Circ. Q (SUS)		1.5	-1.5	-1.5	1.5	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		0.1	-0.1	-1.0	1.0	-0.7	0.7	-0.7	0.7
Circ. Q (TOTAL)		1.6	-1.6	-2.5	2.5	-0.7	0.7	-0.7	0.7
Long. Pm (SUS)		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Long. Pm (OCC)		0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7



Long. Pm(TOTAL)	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Long. P1 (SUS)	0.3	0.3	-0.3	-0.3	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	-0.0	-0.0	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Long. P1(TOTAL)	0.3	0.3	-0.5	-0.5	-0.2	-0.2	-0.2	-0.2
Long. Q (SUS)	2.2	-2.2	-2.2	2.2	0.0	0.0	-0.0	0.0
Long. Q (OCC)	0.2	-0.2	-1.4	1.4	-0.4	0.4	-0.4	0.4
Long. Q (TOTAL)	2.3	-2.3	-3.6	3.6	-0.4	0.4	-0.4	0.4
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	1.1	1.2	1.1	1.2	1.1	1.2	1.1	1.2
Pm (SUS+OCC)	2.6	2.7	2.6	2.7	2.6	2.7	2.6	2.7
Pm+P1 (SUS)	1.9	1.9	0.4	0.4	1.7	1.7	1.7	1.7
Pm+P1 (SUS+OCC)	3.4	3.6	1.4	1.5	3.1	3.2	3.1	3.2
Pm+P1+Q (Total)	5.1	2.7	2.9	4.4	2.5	3.8	2.5	3.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1.19	234.00	Passed
Pm (SUS+OCC)	2.71	280.80	Passed
Pm+P1 (SUS)	1.95	350.99	Passed
Pm+P1 (SUS+OCC)	3.55	421.19	Passed
Pm+P1+Q (TOTAL)	5.06	701.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		5.0	5.1	5.0	5.1	5.0	5.1	5.0	5.1
Circ. Pm (OCC)		6.4	6.5	6.4	6.5	6.4	6.5	6.4	6.5
Circ. Pm(TOTAL)		11.4	11.5	11.4	11.5	11.4	11.5	11.4	11.5
Circ. P1 (SUS)		6.9	6.9	-6.9	-6.9	0.0	0.0	-0.0	-0.0
Circ. P1 (OCC)		0.3	0.3	-4.8	-4.8	-0.9	-0.9	-0.9	-0.9
Circ. P1(TOTAL)		7.1	7.1	-11.7	-11.7	-0.9	-0.9	-0.9	-0.9
Circ. Q (SUS)		5.3	-5.3	-5.3	5.3	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		0.4	-0.4	-3.5	3.5	-9.7	9.7	-9.7	9.7
Circ. Q (TOTAL)		5.7	-5.7	-8.8	8.8	-9.7	9.7	-9.7	9.7
Long. Pm (SUS)		2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Long. Pm (OCC)		3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Long. Pm(TOTAL)		5.7	5.7	5.7	5.7	5.7	5.7	5.7	5.7
Long. P1 (SUS)		3.5	3.5	-3.5	-3.5	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)		0.4	0.4	-2.2	-2.2	-2.3	-2.3	-2.3	-2.3
Long. P1(TOTAL)		3.9	3.9	-5.7	-5.7	-2.3	-2.3	-2.3	-2.3



Long. Q (SUS)	7.4	-7.4	-7.4	7.4	0.0	0.0	-0.0	0.0
Long. Q (OCC)	-1.6	1.6	-7.1	7.1	-5.3	5.3	-5.3	5.3
Long. Q (TOTAL)	5.8	-5.8	-14.4	14.4	-5.3	5.3	-5.3	5.3
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-3.0	-3.0	3.0	3.0
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-4.2	-4.2	4.2	4.2
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	5.0	5.1	5.0	5.1	5.0	5.1	5.0	5.1
Pm (SUS+OCC)	11.4	11.5	11.4	11.5	11.4	11.5	11.4	11.5
Pm+P1 (SUS)	11.9	11.9	1.8	1.8	7.1	7.1	7.1	7.1
Pm+P1 (SUS+OCC)	18.5	18.7	0.3	0.2	12.5	12.5	12.5	12.5
Pm+P1+Q (Total)	24.2	13.0	14.4	14.5	8.8	21.7	8.8	21.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.08	234.00	Passed
Pm (SUS+OCC)	11.54	280.80	Passed
Pm+P1 (SUS)	11.94	350.99	Passed
Pm+P1 (SUS+OCC)	18.66	421.19	Passed
Pm+P1+Q (TOTAL)	24.21	701.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material		SA-516 70
Baseplate Allowable Stress	SBA	137.90 MPa
Baseplate Length	B	150.0000 mm
Baseplate Width	D	150.0000 mm
Baseplate Thickness	BTHK	14.0000 mm
Leg Dimension Along Baseplate Length	d	74.9998 mm
Leg Dimension Along Baseplate Width	b	74.9998 mm
Bolt Circle Diameter	BCD	440.0000 mm
Bolt Material		SA-193 B7
Bolt Allowable Tensile Stress	STBA	137.30 MPa
Bolt Allowable Shear Stress	SBShear	107.87 MPa
Anchor Bolt Nominal Diameter	BOD	20.0000 mm
Number of Anchor Bolts in Tension per Leg	NB	0
Total Number of Anchors Bolt per Leg	NBT	1
Bolt Corrosion Allowance	bca	0.0000 mm
Ultimate 28-day Concrete Strength	FCPRIME	8.750 MPa

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [rb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 1004/(217 • 3)
 = 1.5 MPa. Must be less than 107.9 MPa.

Plate thickness required (1/2"; 13mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be



selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N4 2"

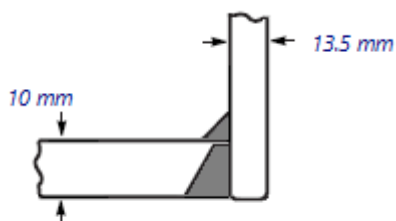
From: 10

Pressure for Reinforcement Calculations	P	0.0542	MPa
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Outside Diameter of Flat Head	D	620.00	mm
Large Diameter of Flat Head	Dl	620.0000	mm
Flat Head Attachment Factor	F	0.30	
Head Finished (Minimum) Thickness	t	12.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	209.5500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N4 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Welded Flat Head, tr [Int. Press]
 $= d \cdot \sqrt{(Z \cdot C \cdot P / (S_v \cdot E))}$ per UG-34 (c)(3)
 $= 600 \cdot \sqrt{(1 \cdot 0.3 \cdot 0.054 / (138 \cdot 1))}$
 $= 6.5138 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.054 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 0.054)$
 $= 0.0112 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	22.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N4 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0112	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 9.5138	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.5138	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(9.51, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.01, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: A

Govrn. thk, tg = 12, tr = 6.51, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.72$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-18 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.05 / 1.96 = 0.028$

Weld Size Calculations, Description: N4 2"



Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.6008 mm

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Input, Nozzle Desc: N1 3"

From: 20

Pressure for Reinforcement Calculations	P	0.0500	MPa
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.00	MPa
Temperature for External Pressure	Tempex	65	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	600.00	mm
Design Length of Section	L	798.0000	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		562.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	15.7480	mm

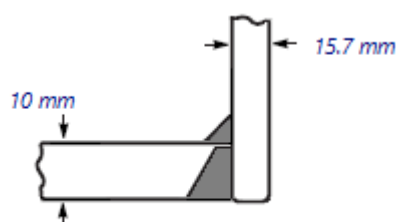
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	210.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1 3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.620 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.05 \cdot 303 / (138 \cdot 1 - 0.6 \cdot 0.05)$
 $= 0.1098$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.05 \cdot 41.1 / (138 \cdot 1 - 0.6 \cdot 0.05)$
 $= 0.0149$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.0539 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	164.4000	mm
Parallel to Vessel Wall, opening length	d	82.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	7.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N1 3".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0539	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.1098	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 8.2578	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(4.5, 4.5)]$
 $= 4.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.05, 4.5)$
 $= 4.5000$ mm

Available Nozzle Neck Thickness = 15.7480 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	12.6,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	332.4 MPa	Passed
Occasional	:	0.1,	Allowable	:	183.5 MPa	Passed
Shear	:	9.8,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.05/5.10 = 0.010

Weld Size Calculations, Description: N1 3"

Intermediate Calc. for nozzle/shell Welds Tmin 3.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	2.1000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 4.8723 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.8723 mm

Input Echo, WRC297 Item 1, Description: N1 3" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	600.000	mm
Shell Thickness	6.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	76.200	mm
Nozzle Thickness	15.7480	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	0.05	MPa
Radial Load	P	-3000.00	N
Circumferential Shear	Vc	2000.00	N
Longitudinal Shear	Vl	2000.00	N
Circumferential Moment	Mc	600000.00	N-mm
Longitudinal Moment	Ml	900000.06	N-mm
Torsional Moment	Mt	0.110E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (6.839) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:



Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 107.696 + 2 \cdot 1.65 \cdot \sqrt{(304.5(6.0 - 3.0))}$$

$$= 207.436 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 609.000 mm
 Nozzle Outside Diameter (d) = 107.696 mm
 Vessel Thickness used (T) = 3.000 mm
 Nozzle Thickness used (t) = 12.748 mm
 T / t = 0.235
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.520$$

Nr/P = 0.063
 Mr/P = 0.122
 M0/P = 0.037
 N0/P = 0.071
 MrD/Mc = 0.277
 NrDL/Mc = 0.092
 M0d/Mc = 0.084
 N0DL/Mc = 0.084
 MrD/Ml = 0.096
 NrDL/Ml = 0.073
 M0D/Ml = 0.028
 N0DL/Ml = 0.092

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	23	23	23	23
Outplane Bending (P)	73	-73	73	-73
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-85	-85	85	85
Outplane Bending (ML)	-156	156	156	-156
Normal Pressure Stress	5	5	5	5
Outplane Stress Summary	-139	26	343	-115

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	20	20	20	20
Inplane Bending (P)	243	-243	243	-243
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-68	-68	68	68
Inplane Bending (ML)	-532	532	532	-532
Inplane Pressure Stress	2	2	2	2
Inplane Stress Summary	-333	243	867	-684

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	20	20	20	20
Shear Stress Summary	24	24	16	16



Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	336	246	868	684

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	23	23	23	23
Outplane Bending (P)	73	-73	73	-73
Outplane Membrane (Mc)	-51	-51	51	51
Outplane Bending (Mc)	-311	311	311	-311
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	2	2	2	2
Outplane Stress Summary	-264	213	462	-306

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	20	20	20	20
Inplane Bending (P)	243	-243	243	-243
Inplane Membrane (Mc)	-57	-57	57	57
Inplane Bending (Mc)	-1030	1030	1030	-1030
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	5	5	5	5
Inplane Stress Summary	-817	755	1356	-1190

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	20	20	20	20
Shear Stress Summary	16	16	24	24

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	818	755	1357	1191

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	23	23	23	23
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-85	-85	85	85
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	-61	-61	109	109



Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	10	-10	10	-10
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-11	-11	11	11
Axial Bending (ML)	-18	18	18	-18
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-18	-2	40	-16

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	5	5	3	3

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	61	61	109	125

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	23	23	23	23
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-51	-51	51	51
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	-27	-27	75	75

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	10	-10	10	-10
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-47	47	47	-47
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-44	31	65	-49

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	3	3	5	5



Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities Cu Cl Du Dl
 Outside Inside Outside Inside

Two * Max Shear Stress 44 59 78 125

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	5	5	5	5	5	5	5
Circ. Pl (SUS)	-62	-62	108	108	-37	-37	77	77
Circ. Q (SUS)	-83	83	229	-229	-787	787	1273	-1273
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-48	-48	88	88	-28	-28	74	74
Long. Q (SUS)	-289	289	775	-775	-238	238	384	-384
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	20	20	20	20	20	20	20	20
Pm (SUS)	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Pm+Pl (SUS)	57.8	57.8	113.4	113.4	33.2	33.2	83.2	83.2
Pm+Pl+Q (Total)	337	245	865	685	819	755	1355	1191

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	138.00	Passed
Pm+Pl (SUS)	113.38	207.00	Passed
Pm+Pl+Q (TOTAL)	1355.59	413.99	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	0	0	0	0	0	0	0
Circ. Pl (SUS)	-62	-62	108	108	-28	-28	74	74
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-11	-11	11	11	-7	-7	7	7
Long. Q (SUS)	-8	8	28	-28	-37	37	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	62.0	62.0	108.0	108.0	28.0	28.0	74.0	74.0
Pm+Pl+Q (Total)	62.4	62.3	108.2	125.3	44.9	58.5	75.4	124.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	108.00	207.00	Passed
Pm+Pl+Q (TOTAL)	125.26	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N2 3"

From: 20

Pressure for Reinforcement Calculations	P	0.0500	MPa
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.00	MPa
Temperature for External Pressure	Tempex	65	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	600.00	mm
Design Length of Section	L	798.0000	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		562.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	15.7480	mm

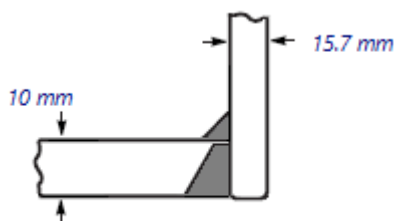
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	210.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2 3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.620 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.05 \cdot 303 / (138 \cdot 1 - 0.6 \cdot 0.05)$
 $= 0.1098$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.05 \cdot 41.1 / (138 \cdot 1 - 0.6 \cdot 0.05)$
 $= 0.0149$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.0539 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	164.4000	mm
Parallel to Vessel Wall, opening length	d	82.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	7.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N2 3".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0539	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.1098	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 4.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 8.2578	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(4.5, 4.5)]$
 $= 4.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.05, 4.5)$
 $= 4.5000$ mm

Available Nozzle Neck Thickness = 15.7480 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	12.9,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	332.1 MPa	Passed
Occasional	:	0.1,	Allowable	:	183.5 MPa	Passed
Shear	:	10.7,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.05/5.10 = 0.010

Weld Size Calculations, Description: N2 3"

Intermediate Calc. for nozzle/shell Welds Tmin 3.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	2.1000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 4.8723 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.8723 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 82.20 mm

Input Echo, WRC297 Item 1, Description: N2 3" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	600.000	mm
Shell Thickness	6.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	76.200	mm
Nozzle Thickness	15.7480	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	0.05	MPa
Radial Load	P	-2000.00	N
Circumferential Shear	Vc	3000.00	N
Longitudinal Shear	Vl	3000.00	N
Circumferential Moment	Mc	600000.00	N-mm
Longitudinal Moment	Ml	900000.06	N-mm
Torsional Moment	Mt	0.110E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (6.839) is < 10

check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 107.696 + 2 \cdot 1.65 \cdot \sqrt{(304.5(6.0 - 3.0))}$
 $= 207.436 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 609.000 mm
 Nozzle Outside Diameter (d) = 107.696 mm
 Vessel Thickness used (T) = 3.000 mm
 Nozzle Thickness used (t) = 12.748 mm
 T / t = 0.235
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.520$$

Nr/P = 0.063
 Mr/P = 0.122
 M0/P = 0.037
 N0/P = 0.071
 MrD/Mc = 0.277
 NrDL/Mc = 0.092
 M0d/Mc = 0.084
 N0DL/Mc = 0.084
 MrD/Ml = 0.096
 NrDL/Ml = 0.073
 M0D/Ml = 0.028
 N0DL/Ml = 0.092

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	48	-48	48	-48
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-85	-85	85	85
Outplane Bending (ML)	-156	156	156	-156
Normal Pressure Stress	5	5	5	5
Outplane Stress Summary	-171	43	310	-98

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	162	-162	162	-162
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-68	-68	68	68
Inplane Bending (ML)	-532	532	532	-532
Inplane Pressure Stress	2	2	2	2
Inplane Stress Summary	-421	318	779	-610

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	5	5	-5	-5
Outplane Shear (Vl)	0	0	0	0



Outplane Shear (Mt)	20	20	20	20
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Shear Stress Summary	26	26	14	14
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Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	424	320	779	610
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	48	-48	48	-48
Outplane Membrane (Mc)	-51	-51	51	51
Outplane Bending (Mc)	-311	311	311	-311
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	2	2	2	2

Outplane Stress Summary	-296	229	430	-290
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	13	13	13	13
Inplane Bending (P)	162	-162	162	-162
Inplane Membrane (Mc)	-57	-57	57	57
Inplane Bending (Mc)	-1030	1030	1030	-1030
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	5	5	5	5

Inplane Stress Summary	-905	829	1268	-1116
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-5	-5	5	5
Torsional Shear (Mt)	20	20	20	20

Shear Stress Summary	14	14	26	26
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	906	830	1269	1117
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-85	-85	85	85
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	0	0	0	0
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Hoop Stress Summary	-69	-69	101	101
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-11	-11	11	11
Axial Bending (ML)	-18	18	18	-18
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-21	0	36	-13
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4

Shear Stress Summary	6	6	3	3
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	69	71	101	114
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-51	-51	51	51
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0

Hoop Stress Summary	-35	-35	67	67
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-47	47	47	-47
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0

Axial Stress Summary	-47	34	62	-46
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
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Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4

Shear Stress Summary	3	3	6	6
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	48	70	71	114

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	5	5	5	5	5	5	5
Circ. Pl (SUS)	-70	-70	100	100	-44	-44	70	70
Circ. Q (SUS)	-108	108	204	-204	-868	868	1192	-1192
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-55	-55	81	81	-36	-36	66	66
Long. Q (SUS)	-370	370	694	-694	-263	263	359	-359
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	5	5	-5	-5	-5	-5	5	5
Shear Q (SUS)	20	20	20	20	20	20	20	20
Pm (SUS)	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Pm+Pl (SUS)	66.8	66.8	106.1	106.1	42.1	42.1	77.6	77.6
Pm+Pl+Q (Total)	425	319	777	611	907	829	1267	1117

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	138.00	Passed
Pm+Pl (SUS)	106.08	207.00	Passed
<i>Pm+PL+Q (TOTAL)</i>	<i>1267.74</i>	<i>413.99</i>	<i>Failed</i>

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	0	0	0	0	0	0	0
Circ. Pl (SUS)	-70	-70	100	100	-36	-36	66	66
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-11	-11	11	11	-7	-7	7	7
Long. Q (SUS)	-12	12	24	-24	-41	41	53	-53
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0



Pm+Pl (SUS)	70.0	70.0	100.0	100.0	36.0	36.0	66.0	66.0
Pm+Pl+Q (Total)	70.5	71.7	100.1	113.2	48.7	70.3	68.8	112.4

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	100.01	207.00	Passed
Pm+Pl+Q (TOTAL)	113.16	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N3 2"

From: 40

Pressure for Reinforcement Calculations	P	0.0500	MPa
Temperature for Internal Pressure	Temp	65	°C
Design External Pressure	Pext	0.00	MPa
Temperature for External Pressure	Tempex	65	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Outside Diameter of Bolted Blind Flange	D	756.60	mm
Head Finished (Minimum) Thickness	t	70.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm

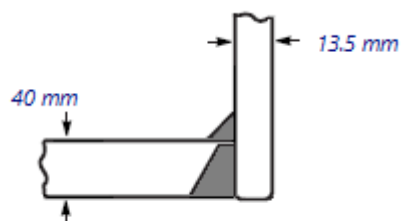
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	209.5500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	70.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N3 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.05 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 0.05)$
 $= 0.0103$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.0443 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	211.7240	mm
Parallel to Vessel Wall	Rn+tn+t	105.8620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N3 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.0443	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 33.9431	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 33.9431	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(33.9, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.04, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 13.5, tr = 0.01, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.00098$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.05 / 1.96 = 0.026$

Weld Size Calculations, Description: N3 2"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.4620	mm
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 279.6008 mm

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